

## A-2. Intercropping

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|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <br>Agriculture Land Conservation | <br>High | <br>Medium |    | <br>M Term |
| Measure Type                                                                                                       | Cost                                                                                      | Preferences by Community                                                                    | Hazards                                                                                                                                                                                                                                                  | Implementation Time of Measures                                                               |

| Descriptions                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |           |      |            |      |            |       |           |
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| Measures                       | <p>Intercropping refers to the practice of growing two or more crops simultaneously on the same piece of land. Intercropping optimizes the limited available space, enhances efficient use of available water and nutrients as well as promotes biodiversity. Intercropping include mixed cropping, companion planting, relay cropping, interseeding, overseeding, underseeding, smother cropping, planting polycultures etc. The crops selected for intercropping are complementary in nature. They have different nutrient requirements, growth habits, or environmental needs, which helps to reduce competition, ensure proper use of available limited resources and improve overall yields.</p> <p>Crops can be planted together without a specific row arrangement, with different crops in different rows or with different crops in different strips. Besides planting different types of crops at the same time, one crop may be planted after another has already started growing depending on their suitable planting period and duration of crop cycle.</p> |       |           |      |            |      |            |       |           |
|                                | Location                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | R-114 | R-99      | R-65 | K-98       | K-76 | K-71       | B-173 | B-77      |
| Applicable Sites               | Smallholder farms, rain-fed areas, degraded slopes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |           |      |            |      |            |       |           |
| Slope                          | Less than 3%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       | 3% to 15% |      | 15% to 30% |      | 30% to 60% |       | Above 60% |
|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |           |      |            |      |            |       |           |
| Design options and performance | <ul style="list-style-type: none"><li>Mixed Cropping – Plants grow without a fixed row arrangement, maximizing biodiversity.</li><li>Companion Planting – Specific plants benefit each other by repelling pests or enhancing nutrient absorption (e.g., maize &amp; legumes).</li><li>Relay Cropping – A second crop is planted before the first is harvested, ensuring continuous soil cover.</li><li>Strip Cropping – Crops are planted in alternate strips, preventing erosion and nutrient depletion.</li><li>Underseeding &amp; Overseeding – Uses cover crops to prevent soil loss and increase organic matter.</li><li>Smother Cropping – Fast-growing crops reduce weed competition and improve soil fertility</li></ul>                                                                                                                                                                                                                                                                                                                                         |       |           |      |            |      |            |       |           |
| Feasibility criteria           | <ul style="list-style-type: none"><li>Choosing crops with similar temperature tolerance and compatible water needs.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |           |      |            |      |            |       |           |

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|                                  | <ul style="list-style-type: none"> <li>Selecting crops based on their pH and nutrients requirements so that different crops can thrive in the same pH condition and not strive for the same soil nutrients.</li> <li>Choosing crops with different root structures to avoid direct competition for nutrients and water in the same soil layers.</li> <li>Ensuring crops that are not prone to similar diseases that could spread between them in close proximity.</li> </ul> |            |               |             |
| <b>Operation and maintenance</b> | Intercropping requires careful timing of field operations and occasionally special interventions to maintain balance between the intercropped species.                                                                                                                                                                                                                                                                                                                       |            |               |             |
| <b>Cost</b>                      | High                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |               |             |
| <b>Benefits</b>                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |               |             |
|                                  | <b>Type</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Low</b> | <b>Medium</b> | <b>High</b> |
|                                  | Canopy Cover Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |               |             |
|                                  | Carbon Sequestration                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |               |             |
|                                  | Crop Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |               |             |
|                                  | Crop Failure Risk Reduction                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |               |             |
|                                  | Ensuring Food Security                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |               |             |
|                                  | Fruit Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |               |             |
|                                  | Forestry Goods Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |               |             |
|                                  | Livestock Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |               |             |
|                                  | Fisheries Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |               |             |
|                                  | Alternative Livelihood Generation                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |               |             |
|                                  | Increase of Income                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |               |             |
|                                  | Soil Erosion Control                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |               |             |
|                                  | Improved Soil Health                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |               |             |
|                                  | Soil Fertility Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |               |             |
|                                  | Sustainable Water Supply                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |               |             |
|                                  | Water Quality Improvement                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |               |             |
|                                  | Water Purification                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |               |             |
|                                  | River Bank Erosion Control                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |               |             |
|                                  | Surface Runoff Reduction                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |               |             |
|                                  | Groundwater Recharge                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |               |             |
|                                  | Pest and Disease Management                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |               |             |
|                                  | Reduction of Pesticides and Fertilizer Use                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |               |             |
|                                  | Ecosystem and Biodiversity Increase                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |               |             |
|                                  | Better Communications                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |               |             |
|                                  | Better Education                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |               |             |
|                                  | Better WASH facilities                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |               |             |
| <b>Implementation Challenges</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |               |             |
|                                  | <b>Type</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Low</b> | <b>Medium</b> | <b>High</b> |

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|                                  | Land unavailable/high land area required                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |  |  |
|                                  | Supply chain disconnection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |  |  |
|                                  | Land right related problems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |  |  |
|                                  | High human investment capital need                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |  |  |
|                                  | Insufficient infrastructures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  |  |
|                                  | Less demand of agroforestry product                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |
|                                  | Environmentally unsustainable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |  |  |
|                                  | Socially unacceptable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |  |
|                                  | Less beneficial                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |  |
| <b>Environmental performance</b> | <ul style="list-style-type: none"> <li>• Intercropping improves soil health and fertility as crops with different root structures prevent soil compaction and promote better water infiltration</li> <li>• Intercropping fosters a diverse agricultural environment, supporting a variety of plant species. The variety of plants in an intercropping system can provide habitats for beneficial insects, birds, and small mammals, promoting local biodiversity.</li> <li>• Intercropping systems often have crops with complementary water needs, which can help optimize water use on the farm. For example, drought-tolerant crops can be paired with more water-demanding crops helping to conserve water.</li> <li>• By disrupting the habitat of pests that favor monocultures, intercropping prevents the buildup of pest populations, thereby reducing the reliance on pesticides and promoting healthier ecosystems.</li> </ul> |  |  |  |
| <b>Remarks</b>                   | This measure will aid landscape restoration in CHT of Bangladesh along with ensuring food security and creating alternative livelihood opportunities.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |  |